

Independent embedding-refinement comparison (1 of 2)

Adjusted Rand index and predicted-cluster Silhouette across all 12 dataset-backbone cases.

Dataset	Backbone	ARI Upstream	ARI GenoRefine	ARI IDEC	Silhouette (SIL) Upstream	Silhouette (SIL) GenoRefine	Silhouette (SIL) IDEC
HP-CB	Scanorama	0.6717	0.4190 ± 0.0355	0.2610 ± 0.0221	0.1729	0.6530 ± 0.0222	0.4025 ± 0.0331
HP-CB	Harmony	0.8213	0.8233 ± 0.0020	0.8027 ± 0.0047	0.2426	0.1796 ± 0.0108	0.0582 ± 0.0143
HP-CB	Seurat	0.8304	0.9154 ± 0.0049	0.8509 ± 0.0407	0.3560	0.2696 ± 0.0311	0.0585 ± 0.0586
HP-CB	Online iNMF	0.5349	0.5904 ± 0.0124	0.3794 ± 0.0181	0.1371	0.2063 ± 0.0249	0.2524 ± 0.0050
Pancreas	Scanorama	0.3896	0.3797 ± 0.0095	0.2548 ± 0.0169	0.1621	0.6576 ± 0.0206	0.3073 ± 0.0196
Pancreas	Harmony	0.6998	0.7194 ± 0.0129	0.7600 ± 0.0572	0.1510	0.1027 ± 0.0063	0.0327 ± 0.0179
Pancreas	Seurat	0.6639	0.7166 ± 0.0229	0.8415 ± 0.0433	0.1962	0.1482 ± 0.0043	0.0308 ± 0.0142
Pancreas	Online iNMF	0.4275	0.5138 ± 0.0106	0.4886 ± 0.0231	0.1368	0.1345 ± 0.0089	0.2721 ± 0.0222
Mouse	Scanorama	0.5403	0.5368 ± 0.0223	0.4661 ± 0.0027	0.2359	0.7436 ± 0.0346	0.4640 ± 0.0642
Mouse	Harmony	0.5042	0.5067 ± 0.0016	0.4982 ± 0.0058	0.2723	0.2449 ± 0.0039	0.1394 ± 0.0064
Mouse	Seurat	0.5143	0.5076 ± 0.0078	0.4948 ± 0.0065	0.2797	0.2561 ± 0.0051	0.1458 ± 0.0077
Mouse	Online iNMF	0.5066	0.4607 ± 0.0191	0.4451 ± 0.0102	0.4187	0.4985 ± 0.0098	0.6343 ± 0.0818

Upstream is a fixed embedding. GenoRefine and IDEC are mean ± sample SD across five algorithmic seeds; seeds are not biological replicates.

Independent embedding-refinement comparison (2 of 2)

Batch mixing (iLISI) and local label purity across all 12 dataset-backbone cases.

Dataset	Backbone	iLISI Upstream	iLISI GenoRefine	iLISI IDEC	Local label purity Upstream	Local label purity GenoRefine	Local label purity IDEC
HP-CB	Scanorama	0.1250	0.1791 ± 0.0062	0.2523 ± 0.0084	0.9627	0.9495 ± 0.0079	0.8469 ± 0.0257
HP-CB	Harmony	0.1773	0.1932 ± 0.0011	0.2143 ± 0.0024	0.9534	0.9475 ± 0.0012	0.8991 ± 0.0032
HP-CB	Seurat	0.2510	0.2678 ± 0.0004	0.2699 ± 0.0018	0.9498	0.9489 ± 0.0005	0.9348 ± 0.0015
HP-CB	Online iNMF	0.1386	0.1339 ± 0.0009	0.1579 ± 0.0078	0.9498	0.9502 ± 0.0008	0.9394 ± 0.0027
Pancreas	Scanorama	0.0195	0.0533 ± 0.0079	0.1515 ± 0.0142	0.9531	0.9533 ± 0.0023	0.8499 ± 0.0232
Pancreas	Harmony	0.1211	0.1203 ± 0.0027	0.1801 ± 0.0066	0.9470	0.9387 ± 0.0021	0.8726 ± 0.0055
Pancreas	Seurat	0.2652	0.2698 ± 0.0017	0.2814 ± 0.0026	0.9502	0.9459 ± 0.0006	0.8914 ± 0.0037
Pancreas	Online iNMF	0.0184	0.0175 ± 0.0006	0.0412 ± 0.0044	0.9289	0.9324 ± 0.0010	0.9262 ± 0.0023
Mouse	Scanorama	0.3441	0.4887 ± 0.0088	0.5678 ± 0.0135	0.6992	0.6096 ± 0.0028	0.5510 ± 0.0057
Mouse	Harmony	0.3839	0.4022 ± 0.0022	0.4453 ± 0.0057	0.6615	0.6593 ± 0.0004	0.6329 ± 0.0011
Mouse	Seurat	0.4832	0.5065 ± 0.0032	0.5350 ± 0.0047	0.6659	0.6632 ± 0.0008	0.6371 ± 0.0008
Mouse	Online iNMF	0.2824	0.2197 ± 0.0052	0.1089 ± 0.0126	0.5916	0.5875 ± 0.0010	0.5724 ± 0.0021

Upstream is a fixed embedding. GenoRefine and IDEC are mean ± sample SD across five algorithmic seeds; seeds are not biological replicates.